

★ WEEK 12 · PART 1 · PRACTICE

Short-Rate Models, Worked by Hand

Vasicek's Negative-Rate Catch, and CIR's Positive Floor

OU for interest rates, and the square-root fix

Part 1's short-rate models, with the arithmetic shown. We compute the probability Vasicek assigns to negative rates, then check CIR's Feller condition and see how its square-root diffusion vanishes at zero to keep rates positive.

1P

PART 1 · APPLICATIONS

Two Rate Models, by Hand

Case A: Vasicek as OU for the short rate - its stationary distribution, and the probability it goes negative. Case B: CIR's square-root diffusion and the Feller condition that keeps rates strictly positive. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Vasicek is OU for the short rate

The same mean-reversion SDE from Week 11 - rates pulled home to a long-run level θ at speed κ .

2

Vasicek's catch is negative rates

Its constant volatility makes the stationary distribution Gaussian, so it puts real probability below zero.

3

CIR fixes it with a square root

Scaling the noise by $\sqrt{r}$ makes it vanish at zero. The Feller condition guarantees rates stay strictly positive.

Everything here is a calculator exercise: one stationary distribution, one Feller check.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a rate-model SDE.
- Compute a probability or a condition by hand.
- Then read what it means for rates.

What you should watch

- How constant vol lets rates go negative.
- How the Feller condition guards positivity.
- How the square-root noise dies at zero.

Vasicek's Negative-Rate Catch

A Gaussian tail below zero

Vasicek is just the OU process applied to the short rate. That makes its long-run distribution a Gaussian centered on the mean-reversion level - which means it assigns real probability to rates below zero. With a modest mean and a typical volatility, we compute exactly how much

The parameters

- Reversion speed $\kappa = 0.30$.
- Long-run level $\theta = 2.5\%$.
- Volatility $\sigma = 1.5\%$.

What we'll do by hand

- Find the stationary mean and sd.
- Standardize zero into a z-score.
- Read off the probability of a negative rate.

Vasicek and Its Stationary Law

OU, with a Gaussian equilibrium

$$\begin{aligned} \text{Vasicek: } dr &= \kappa(\theta - r) dt + \sigma dW && (\text{OU, applied to the short rate}) \\ \text{stationary: } r_\infty &\sim N\left(\theta, \frac{\sigma^2}{2\kappa}\right) && (\text{constant } \sigma \Rightarrow \text{Gaussian}) \end{aligned}$$

The dynamics

- Identical to OU - a pull toward theta.
- The diffusion is a constant sigma.
- That constant is the source of the problem.

The equilibrium

- The stationary law is a normal distribution.
- Centered at theta, with a fixed spread.
- And a normal has tails on both sides.

The Stationary Mean and SD

Where rates settle, and how wide

$$\text{mean} = \theta = 2.5\%, \quad \text{sd} = \frac{\sigma}{\sqrt{2\kappa}} = \frac{1.5\%}{\sqrt{0.60}} = \mathbf{1.94\%}$$

The mean

- Rates settle around $\theta = 2.5\%$.
- That's the long-run reversion level.
- The center of the distribution.

The spread

- The sd is sigma over root two-kappa.
- That works out to 1.94%.
- Wide enough to reach below zero.

The Probability of Negative Rates

Standardize zero, and read the tail

$$P(r < 0) = \Phi\left(\frac{0-2.5\%}{1.94\%}\right) = \Phi(-1.29) = \mathbf{9.8\%} \quad (\text{Vasicek's negative-rate catch})$$

The calculation

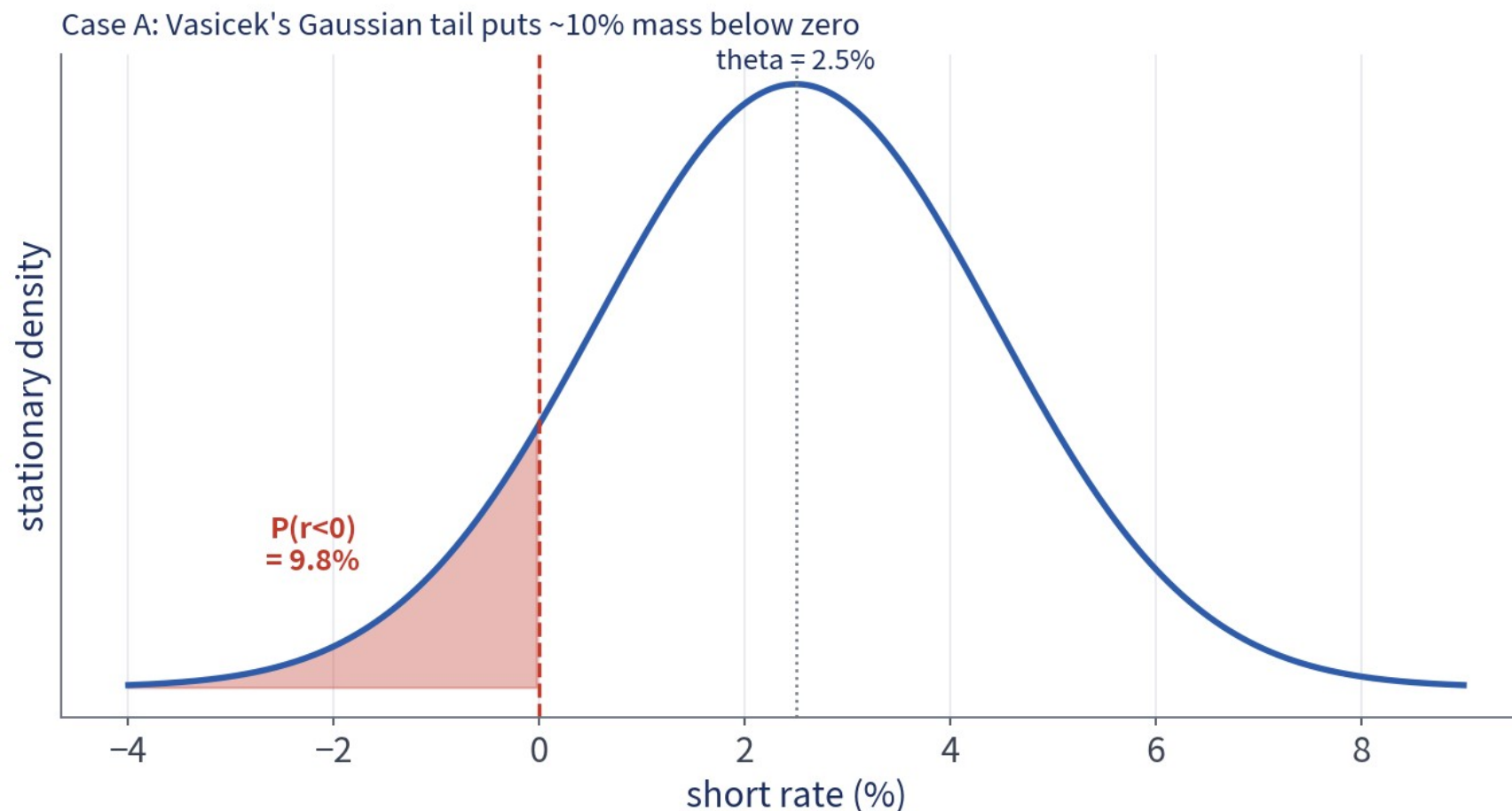
- Zero is 1.29 sd below the 2.5% mean.
- The normal CDF at minus-1.29 is 0.098.
- So about a 10% chance of negative rates.

The verdict

- That's not negligible - it's one year in ten.
- The constant-vol assumption did this.
- This is Vasicek's structural weakness.

A Tenth of the Mass Below Zero

The stationary distribution is a bell curve centered at 2.5%, but its left tail crosses zero. The shaded region - about 10% of the probability - is where Vasicek says rates are negative. Historically awkward, and the direct cost of a constant volatility



Why the Catch Matters

Constant vol has a price

The problem

- A Gaussian has no floor.
- The lower the mean, the more mass below zero.
- Bond prices from it can misbehave.

When it's tolerable

- Negative rates are no longer purely theoretical.
- Vasicek stays analytically clean and tractable.
- For many uses the 10% tail is acceptable.

CIR and the Feller Condition

A square root keeps rates positive

CIR keeps the same mean reversion but scales the noise by the square root of the rate. As the rate falls toward zero, the volatility shrinks with it, so the randomness can't push it across the boundary. The Feller condition tells you exactly when this guarantee holds. We check it by hand

The model

- Same reversion: $\kappa = 0.30$, $\theta = 3\%$.
- But the diffusion is σ times $\sqrt{r}$.
- Two candidate volatilities to test.

What we'll do by hand

- Compute $2\kappa\theta$.
- Compare it to σ^2 for each.
- See the diffusion vanish as r goes to zero.

CIR and Feller

Where the square root earns its keep

$$\text{CIR: } dr = \kappa(\theta - r) dt + \sigma\sqrt{r} dW \quad (\text{square-root diffusion})$$

Feller: $2\kappa\theta \geq \sigma^2 \Rightarrow$ the rate stays strictly positive

The diffusion

- The noise is sigma times root-r.
- It shrinks as the rate approaches zero.
- That's what protects the boundary.

The condition

- Feller: two-kappa-theta at least sigma-squared.
- Then the rate never touches zero.
- Reversion must beat the noise.

Checking the Feller Condition

Two volatilities, two verdicts

$$2\kappa\theta = 2(0.30)(0.03) = 0.018$$

$$\sigma = 0.10 : \sigma^2 = 0.010 \leq 0.018 \Rightarrow \text{HOLDS (r \> 0)}$$

$$\sigma = 0.15 : \sigma^2 = 0.0225 > 0.018 \Rightarrow \text{VIOLATED (r can hit 0)}$$

With two-kappa-theta at 0.018, a volatility of 0.10 satisfies Feller and rates stay strictly positive; 0.15 violates it, and the rate can reach zero.

The Noise Dies at the Floor

Why zero is a wall, not a door

$$\sigma\sqrt{r} : \quad r = 3\% \rightarrow 0.017, \quad r = 0.5\% \rightarrow 0.007, \quad r \rightarrow 0 \Rightarrow 0 \quad (\text{the noise dies at the floor})$$

The mechanism

- At 3%, the diffusion is a healthy 0.017.
- At 0.5%, it's already halved to 0.007.
- At zero, it's exactly zero.

The consequence

- With no noise at the boundary, only drift acts.
- And drift points up, back toward theta.
- So the rate is pushed away from zero.

One Dips Below, One Bounces Off

Driven by the same shocks, the Vasicek path (red) plunges below zero when its constant volatility overwhelms the weak upward drift, while the CIR path (green) has its noise fade near the floor and bounces off zero. Same reversion, different diffusion, opposite behavior at the boundary.

Case B: constant vol lets rates go negative; sqrt diffusion holds the floor



Why CIR for Rates

Positivity, built into the diffusion

What CIR buys

- Rates that respect a zero floor (under Feller).
- Volatility that rises with the level - realistic.
- Still tractable, with closed-form bond prices.

The trade-off

- The math is a bit heavier than Vasicek.
- The Feller condition can bind on calibration.
- But no embarrassing negative rates.

Same Reversion, Different Diffusion

The one line that separates them

Vasicek and CIR share the exact same mean-reversion drift; they differ only in the noise term. Vasicek's constant sigma gives a clean Gaussian - and a negative tail. CIR's sigma-times-root-r gives a floor at zero - at the cost of a Feller condition. The whole choice lives in the diffusion

Vasicek

- Constant volatility.
- Gaussian, negative tail (10% here).
- Simplest, cleanest, but can go below zero.

CIR

- Volatility scales with root-r.
- Positive under Feller ($2\kappa\theta \geq \sigma^2$).
- Realistic and positive, slightly heavier.

Reproduce the Tail and the Floor

Direct Claude Code to check your arithmetic

GOAL Verify Vasicek's negative-rate probability and CIR's Feller check, then simulate both.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - Vasicek:

- * from $\kappa=0.30$, $\theta=0.025$, $\sigma=0.015$,
print the stationary $sd = \sigma/\sqrt{2\kappa}$
and $P(r<0)$ via the normal CDF.

CASE B - CIR:

- * with $\kappa=0.30$, $\theta=0.03$, print $2\kappa\theta$
and check Feller for $\sigma = 0.10$ and 0.15 ;
- * print $\sigma\sqrt{r}$ at $r = 3\%$, 0.5% , 0 .

Then simulate Vasicek and CIR paths and count
how often each goes negative.

Match every printed number to the slides.

Check against the slides

- stationary $sd = 1.94\%$.
- $P(r<0) = 9.8\%$.
- $2\kappa\theta = 0.018$.
- Feller: 0.10 holds, 0.15 fails.

Deliverable: `part1_practice.py` whose printed numbers match the negative-rate tail and the Feller check.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why can Vasicek produce negative rates?

Constant vol makes the law Gaussian

Q2 What is the Feller condition?

2-kappa-theta at least sigma-squared - rates stay positive

Q3 How does CIR keep rates positive?

The root-r diffusion vanishes at zero

All three come from Vasicek's stationary law and CIR's Feller condition.

Three Pitfalls to Avoid

Where rate models trip people

Common mistakes

- "Vasicek can't give negative rates."
- "CIR is always positive."
- "CIR and Vasicek differ in their drift."

The reality

- Its Gaussian tail puts real mass below zero - 10% here.
- Only if Feller holds; violate it and zero is reachable.
- The drift is identical - only the diffusion differs.

Two Calculations to Remember

Part 1 applications in one screen

1

Vasicek assigns real mass to negative rates

As OU for the short rate, its stationary law is normal - mean 2.5%, sd 1.94%. Standardizing zero gives a z of -1.29 and a probability of 9.8% that rates are negative. That's the price of a constant volatility.

2

CIR's square root holds the floor

With two-kappa-theta = 0.018, a volatility of 0.10 satisfies Feller and rates stay strictly positive, while 0.15 violates it. The diffusion sigma-root- r falls from 0.017 at 3% to zero at the boundary - so the noise dies exactly where it would do harm.

3

Same reversion, different diffusion

Vasicek and CIR share the mean-reversion drift and differ only in the noise. Constant vol gives simplicity and a negative tail; root- r gives positivity and a Feller condition. The whole modeling choice is one term.

Next — Part 2 practice: from the short rate to the whole term structure of bond yields.